



Mediacrest Training College

Nairobi, Kenya

Advanced Data Analytics & Machine Learning

8-Week Advanced Training Program

Capstone Project Documentation

Machine Learning Capstone Project

Customer Churn Prediction & Predictive Pipelines

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“A model is only as valuable as the business decisions it informs. Analytics must bridge the gap between raw data and strategic action.”

Project Scenario: Advanced Machine Learning for Churn Prediction

You have been hired as the **Lead Machine Learning Engineer** for a rapidly growing Kenyan digital lifestyle brand. Following the initial strategic analysis of the business, the executive team has mandated the development of a production-ready, highly accurate machine learning system to predict subscription churn.

The product and retention teams are struggling with high customer acquisition costs and need an automated, data-driven early warning system to identify subscribers at risk of cancellation before they leave.

You are provided with the dataset:

- `churn_detection_data.csv`: Contains customer demographics, account behavior, transaction metrics, complaint history, and a binary target variable (**Churn**) indicating subscription cancellation.

Your mission is to design, engineer, train, and evaluate a comprehensive suite of machine learning models using robust preprocessing pipelines, ultimately selecting the best algorithm to deploy into the company's retention strategy.

Project Requirements

This capstone requires you to execute an end-to-end machine learning pipeline. You will compare 12 distinct algorithms ranging from linear models to advanced gradient boosting frameworks. You must complete all **12 Tasks** outlined in this document.

Academic Integrity Code Hints

This document intentionally contains **no Python code hints**. You are expected to independently research, write, and debug the Python code required to execute these tasks using libraries such as `pandas`, `scikit-learn`, `xgboost`, `catboost`, `matplotlib`, and `seaborn`.

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1 Machine Learning Pipeline & Tasks

1.1 Data Preparation & Exploratory Analysis

Before modeling, the data must be understood, cleaned, and transformed.

Task 1: Data Importation and Initial Audit

1. Load `churn_detection_data.csv` into your analytical environment.
2. Perform an initial audit: display dimensions, data types, and summary statistics.
3. Identify missing values, duplicates, and structural anomalies.
4. Apply appropriate cleaning strategies (e.g., handling missing numerical values with median/mean, and addressing categorical missingness).

Task 2: Data Wrangling and Preprocessing

1. Identify and handle outliers in continuous variables like `Average_Balance` and `Account_Tenure_Months`.
2. Analyze the target variable `Churn` for class imbalance. Document the churn rate.
3. Separate features (X) and the target (y).

Task 3: Exploratory Data Analysis (EDA) & Visualization

1. Plot the distribution of customer churn across key demographics (Age bins, Gender, Education, Marital Status, Employment Status).
2. Visualize the relationship between numerical features (e.g., `Number_of_Products`, `Complaint_Resolution_Rate`) and churn.
3. Generate a correlation heatmap for numerical features to identify potential multi-collinearity.

1.2 Feature Engineering & Pipeline Construction

Robust pipelines ensure reproducibility and prevent data leakage.

Task 4: Feature Engineering and Selection

1. Create derived metrics (e.g., `Balance_per_Tenure_Month`, interaction terms between `Number_of_Complaints` and `Complaint_Resolution_Rate`).
2. Apply feature selection techniques (e.g., Mutual Information, or tree-based feature importance) to reduce dimensionality if necessary.

Task 5: Pipeline Construction & ColumnTransformer

1. Construct a `ColumnTransformer` to apply distinct preprocessing steps to numerical and categorical features.
2. **Numerical Pipeline:** Imputation (if needed) → Scaling (e.g., `StandardScaler` or `RobustScaler`).
3. **Categorical Pipeline:** Imputation → Encoding (e.g., `OneHotEncoder` or `OrdinalEncoder`).
4. Chain these transformations with your estimators using scikit-learn `Pipeline` objects to prevent data leakage during cross-validation.

The Importance of Pipelines

Using `Pipeline` and `ColumnTransformer` ensures that the exact same preprocessing steps applied during training are automatically applied to test data or live inference requests, eliminating a major source of deployment bugs.

1.3 Baseline & Regularized Linear Models

Establishing a baseline and understanding feature linearity.

Task 6: Baseline Model - Binary Logistic Regression

1. Build a Binary Logistic Regression model using your pipeline.
2. Interpret the coefficients and calculate the Odds Ratios (OR) for key variables.
3. Evaluate using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

Task 7: Regularized Models - Ridge, Lasso, Elastic Net

1. Implement Ridge (L2), Lasso (L1), and Elastic Net regressions.
2. Use `GridSearchCV` or `RandomizedSearchCV` within your pipeline to tune regularization strength (`C` or `alpha`) and the L1 ratio.
3. Use the Lasso model to identify and drop features with coefficients shrunk exactly to zero.

Elastic Net Penalty

The Elastic Net combines L1 and L2 penalties:

$$\text{Loss} = \text{MSE} + \alpha \left(\rho \sum_{i=1}^n |\beta_i| + \frac{1-\rho}{2} \sum_{i=1}^n \beta_i^2 \right)$$

Where ρ controls the mix between Lasso and Ridge.

1.4 Distance, Margin, and Tree-Based Models

Capturing non-linear relationships and complex interactions.

Task 8: Instance-Based and Margin Models - KNN & SVM

1. Build a K-Nearest Neighbors (KNN) classifier. Tune `n_neighbors` and distance metrics.
2. Build a Support Vector Machine (SVM) classifier. Experiment with linear and RBF kernels, tuning the `C` and `gamma` parameters.
3. **Critical:** Verify that your pipeline's scaling step is functioning correctly, as both KNN and SVM are highly sensitive to unscaled features.

Task 9: Tree-Based Ensembles - Decision Tree, Random Forest, Extra Trees

1. Train a single Decision Tree. Visualize the tree structure and analyze its tendency to overfit.
2. Build a Random Forest and an Extra Trees classifier. Tune `n_estimators`, `max_depth`, and `min_samples_split`.
3. Extract and plot the `feature_importances_` for both ensemble models. Which demographic or account factors drive churn the most?

1.5 Advanced Boosting Algorithms

State-of-the-art gradient boosting for tabular data.

Task 10: Boosting Algorithms - XGBoost, CatBoost, AdaBoost

1. Implement **XGBoost**: Tune learning rate, max depth, and subsample ratios. Handle class imbalance using `scale_pos_weight`.
2. Implement **CatBoost**: Leverage its native categorical feature handling capabilities (pass categorical indices directly). Compare its training time and performance against XGBoost.
3. Implement **AdaBoost**: Tune the number of estimators and learning rate.

Kenyan Market Context

In the Kenyan digital market, acquiring a new subscriber via mobile money integrations or social media ads can cost 3x to 5x more than retaining an existing one. Furthermore, telecom and digital subscription churn is highly sensitive to network reliability and mobile money transaction failures. Your model's ability to correctly flag at-risk users (Recall) directly impacts the bottom line.

1.6 Model Evaluation & Business Strategy

Synthesizing results into actionable intelligence.

Task 11: Comprehensive Model Evaluation and Comparison

1. Compile a comparison table of all 12 models evaluating Accuracy, Precision, Recall, F1-Score, and ROC-AUC on the hold-out test set.
2. Plot the ROC curves for all models on a single chart for visual comparison.
3. Plot Precision-Recall (PR) curves, which are more informative for imbalanced churn datasets.
4. Select the **Champion Model** based on the business requirement of maximizing Recall while maintaining acceptable Precision.

Handling Imbalanced Data

If your champion model struggles with Recall, consider implementing SMOTE (Synthetic Minority Over-sampling Technique) or adjusting the `class_weight='balanced'` parameter within your pipeline's estimator step.

Task 12: Business Strategy & Policy Memo

Synthesize your machine learning findings into a 2-page strategic memo addressed to the Executive Team. Your memo must include:

1. **Champion Model Selection:** Justify why the chosen model (e.g., CatBoost or XGBoost) is best for production (consider accuracy, inference speed, and interpretability).
2. **Key Churn Drivers:** Based on feature importance and SHAP values (if computed), what are the top 3 drivers of churn?
3. **Targeted Retention Policy:** Propose a data-driven intervention strategy. How should the customer success team use the model's predictions to allocate retention budgets (e.g., offering discounts to high-risk, high-value customers)?
4. **Model Monitoring:** Briefly outline how the team should monitor for data drift or model decay post-deployment.

2 Deliverables & Submission Guidelines

2.1 Required Outputs

1. **Jupyter Notebook:** A clean, modular, and well-commented notebook containing all Python code, pipeline definitions, and outputs for Tasks 1 through 11.
2. **Executive Memo (PDF):** The strategic business memo requested in Task 12.
3. **Presentation Deck:** A 5-slide summary deck highlighting the pipeline architecture, model comparison (ROC/PR curves), and strategic recommendations.

2.2 Evaluation Rubric

- **Technical Rigor (40%):** Correct implementation of pipelines, cross-validation, hyperparameter tuning, and appropriate evaluation metrics for imbalanced data.
- **Code Quality (20%):** Clean, modular, and well-documented Python code. Proper use of `ColumnTransformer` and `Pipeline`.
- **Business Acumen (30%):** Depth of insights, correct interpretation of model metrics (e.g., prioritizing Recall), and relevance of the strategic recommendations.
- **Communication (10%):** Clarity of the executive memo and presentation.

The Prevalence Trap Metric Selection

Do not rely on Accuracy for imbalanced churn datasets. If only 15% of customers churn, a dummy model predicting "No Churn" for everyone achieves 85% accuracy but is entirely useless. Focus on **ROC-AUC**, **F1-Score**, and **Recall**.